

Mini Project: Color Detector Using OpenCV

1. Objective:

Detect and display the BGR color values of pixels clicked in a live webcam feed using Python and OpenCV.

2. Tools Used:

- Python
- OpenCV
- Visual Studio Code

3. Description of Steps:

- Opened webcam feed using `cv2.VideoCapture(0)`
- Created a mouse event callback using `cv2.setMouseCallback()`
- On mouse click, read BGR values of the pixel
- Displayed BGR values both on the console and overlaid on the webcam window

4. Output:

- Webcam feed displays a colored rectangle and BGR values at top-left when a user clicks
- BGR values also printed in the terminal

5. Learning Outcome:

Understood how to work with real-time webcam data and handle mouse interaction in OpenCV.
Learned how to manipulate image data and display dynamic overlays.